Ab-initio Molecular Dynamics

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August 6, 2026

Macro-world: Thermodynamic Conjugated Variables

The Concept

State variables come in pairs:

- one extensive (depends on the size of the system, like the number of atoms)
- one intensive (independent of size, like temperature)

Only one of each can be controlled.

The typical pairs are

- $N \leftrightarrow \mu$: Number of particles (N) vs. Chemical Potential (μ). In closed-box MD, we fix N.
- $V \leftrightarrow p$: Volume (V) vs. Pressure (p). You either lock the box size, or you let it breathe to maintain a target pressure.
- $E \leftrightarrow T$: Total Energy (E) vs. Temperature (T). Either energy is conserved, or a heater sets an average temperature.

Micro-world: Statistical Ensembles

Goal

Translate the macroscopic variables into the standard simulation setups (Ensembles).

What is an Ensemble? A collection of all possible microscopic arrangements (positions and momenta of atoms) that satisfy a macroscopic thermodynamic state.

Imagine taking a million photographs of a chemical system at the exact same temperature and pressure. Every photo shows a slightly different microscopic arrangement of atoms. The complete collection of all those possible photos is the **Ensemble**.

Micro-world: Statistical Ensembles

The Simulation Trick (Time vs. Statistics):

- In MD, we do not run a million boxes. We run one box.
- How do we get the ensemble? over a long time, the system naturally wanders through all those different "photograph" arrangements.
- The Golden Rule: Averaging a property (like E or p) over a long time in a single run is equivalent to averaging across the ensemble.

Warning: If you don't run the simulation long enough, or if the atoms get stuck in a state, you are only seeing a tiny fraction of the "photos." Your statistics will be completely wrong!

The microcanonical ensemble

Microcanonical (NVE): The Isolated System

- **Fixed**: N, V, E.
- **Physics**: Like a perfectly insulated thermos. Energy is conserved, but Temperature T fluctuates naturally as bonds stretch and compress.
- **Use:** The gold standard for diffusion, but practically difficult because it's hard to hit a specific target temperature. (See more on this later)

The Canonical ensemble

Canonical (NVT): The Closed Flask in a Water Bath

- **Fixed**: N, V, T.
- **Physics**: The system can exchange heat with an infinite external bath. Energy E fluctuates.
- **Use:** The most common production ensemble for solid-state/materials science.

The Isothermal-Isobaric ensemple

Isothermal-Isobaric (NpT): The Flexible Balloon

- **Fixed**: N, p, T
- **Physics**: The box volume V fluctuates to maintain a target pressure p (usually 1 atm).
- **Use**: Crucial for phase transitions (melting) and finding the correct equilibrium density of a material

Bridging the Gap (Why Thermostats?)

Goal

Explain how we actually force the system to hold a constant T (since standard $F = ma$ naturally conserves E, not T).

- **The Problem**: The *Equipartition Theorem* relates T to the kinetic energy of the atoms: $\langle K \rangle = \frac{3}{2} N k_B T$. To control T, we have to artificially modify the velocities of the atoms.
- **Thermostat**: An algorithm that acts as a heat bath, adding or removing K from the system to maintain the target T.
- **Warning**: Because thermostats artificially change atomic velocities, they can ruin real physics if chosen poorly!

The Thermostat Cheat Sheet

- **Velocity Rescaling**
  - *How it works*: Brute-force multiplies all velocities by a scaling factor to force the temperature to the target.
  - *Pro/Cons*: Do not use. It violates statistical mechanics and suppresses natural thermal fluctuations.
- **Stochastic (Langevin / Andersen)**
  - *How it works*: Adds artificial friction and random "kicks" to the atoms (Brownian motion).
  - *Pro/Cons*: Good for equilibration (even melting a crystal). Terrible for transport because the friction slows down diffusion
- **Extended System (Nosé-Hoover)**
  - *How it works*: Adds a fictitious "thermal mass" to the equations of motion. The system smoothly exchanges energy with this mass.
  - *Pro/Cons*: Preserves the true time-evolution of the atomic trajectories. The absolute best choice for production runs and calculating observables

The Small Box Problem ($1/\sqrt{N}$)

- **Macroscopic vs. Microscopic**: In a real test tube (10^{23} atoms), fluctuations are zero. In VASP (100 atoms), fluctuations are roughly 10%.
- **Instantaneous vs. Average**: You cannot trust a single step of an MD trajectory to give you a thermodynamic property.
- **Pressure Shock**: Instantaneous pressure in MD will look completely wrong (fluctuating by gigapascals). This is normal! Only the time-averaged pressure $\langle p \rangle$ corresponds to the laboratory pressure

The Rule of Thumb: The state variables (T, p, V, E) are only meaningful after they have been averaged over thousands of femtoseconds.

Starting an MD: The Temperature Crash

- **Step 0: The Perfect Crystal**
  - Atoms start at their perfect geometry (the bottom of the energy well).
  - Potential Energy: $V = 0$ (defined for ground state).
  - Total Energy is entirely kinetic: $E = K_{initial}$
- **Step 1: Climbing the Well**
  - As atoms move, bonds stretch. V increases.
  - To conserve total energy ($E = K + V$), K must drop.
- **The Physics (Virial Theorem)**:
  - For harmonic bonds at equilibrium, energy splits equally: $\langle K \rangle = \langle V \rangle$.
  - Since E was initially just $K_{initial}$, the final state is:

$$E = \langle K \rangle + \langle V \rangle = 2\langle K \rangle$$

  - Because $2\langle K \rangle = K_{initial}$, $\implies$ $\langle T \rangle$ drops to $T_0/2$

The Role of the Thermostat (Equilibration vs. Production)

1. Equilibration Run (Filling the Well)
   - **The Goal**: Pump the "missing" energy back into the system.
   - Because the atoms absorbed half $V_{initial}$, the thermostat must add that energy back until $\langle T \rangle = T_0$ and $E = 2K_{initial}$.
   - **Key rule**: Never measure physical properties here. The system is fundamentally changing its state and energy.
2. **Production Run (Measuring Reality)**
   - **The Goal**: Observe the system's natural dynamics.
   - Once the potential wells are "full" ($\langle K \rangle = \langle V \rangle$ at T_0), we switch to a gentle thermostat (Nosé-Hoover) or turn it off entirely (NVE).
   - Now, we can safely *average* diffusion, radial distribution functions, and vibrational spectra.

Session structure and a warning

- The simulation will take a lot of time. So you may give up, it is ok.
- There is a `spoiler` subfolder at each folder. It is related to what you should achieve

WARNING

The simulation is strongly sacrificing accuracy in favor of speed. The `spoiler` folder do not suffer from these innacuracies.

The diffusion of BCC Li

Note: everything is at the SimMat page, `https://github.com/simmat-uchile/MiniSchool_SimMat_2026`
We are going to use VASP to calculate the diffusion coefficient Li, at 600 K. The lesson has some steps, intentional mistakes, etc.
First thing first, we need a Li crystal to start with.

- Search for `Materials project Li`
- Pick the cubic lattice (BCC)
- Within the animated crystal view, click `Export as`→`VASP input`
- extract these files

Main input files

- Inspect the Materials project's input
- Go to the (github) folder `0_bulk_Li`
- There are 4 files:
  - `INCAR`: What to do
  - `POSCAR`: Geometry
  - `KPOINTS`: Reciprocal-space stuff
  - `POTCAR`: Pseudopotentials
- Read this `INCAR` it is a more human-friendly version of the one in materials project.

Creating a Supercell

- install ASE in your laptop `https://ase.gitlab.io/ase`
  `pip install ase`
- For a MD we need a supercell, let's create a $2 \times 2 \times 2$ cell. 54 atoms.
- This is **too small** for a proper MD, but it is fast.
- Go to the folder `1_md_equilibrium`, it has couple of scripts, `INCAR`, `POTCAR`, `POSCAR-unitcell`, but no `POSCAR`.
- The file `POSCAR-unitcell` is just the unit cell's `POSCAR`, as downloaded from materials project. Run
  `python create-supercell.py`
- Now you have a `POSCAR`, inspect it. Also inspect `create-supercell.py`, changing the cell size is trivial. Copy it to leftraru.
- If anything fails, just do:
  `cp spoiler/POSCAR .`

Creating a Supercell

How to make such a supercell script? Ask your favorite chatbot:

```
"Write a Python script using the ase library to
generate a VASP supercell.  Requirements:
1.  Read an input file named exactly POSCAR-unitcell.
2.  Expand it to create a 2x2x2 supercell.
3.  Save the output as POSCAR (VASP format).
4.  Print the initial and final number of atoms to
verify."
```

Running a equilibration MD

- go to `1_md_equilibrium` in leftraru.
- Make sure you have the files `INCAR`, `POSCAR`, `POTCAR`, `KPOINTS`
- Execute `sbatch run.sh`
- Run `squeue`, the status of your calculation
- Run `bash plot.sh` to inspect your results in realtime.
- Is your system thermalized? What about pressure?
- Notice the drop in T.

Running a equilibration MD 2

- You can retrieve the files from the server and watch the trajectory.
- I recommend you using `ovito`, `https://www.ovito.org/`
- `Vesta` is great for static visualization `https://jp-minerals.org/vesta/en/`
- Mind: if you want, go to the `spoiler`, it has the required files.

Running a equilibration MD 3

- The system was not even close to equilibrium!
- Let's increase the time to 4 ps (4000 steps)
- `cd ../2_md_equilibrium`
- It takes a long time, even for such a small cell.
- Let's accelerate the process. Go to `spoiler`.
- Plot the thermodynamical variables. Are they constant? What about the fluctuations? Their average?
- In your laptop, watch the dynamics.
- If you need to abort a calculation `qdel 23235665` (change the number by your job ID).

Running a Production MD

- `cd 3_md_production`
- Copy your equilibrated `CONTCAR`:
  `cp ../2_md_equilibrium/CONTCAR POSCAR`
- Inspect the *INCAR* file. What has changes?
- Run the calculation
- Again it will take too long. Let's analyze the `spoiler` folder.
- Plot the thermodynamical variables. What could you conclude?

What is Diffusion? From Macroscopic to Microscopic

The Macroscopic View (Chemistry):
Mass transport driven by concentration gradients. It is the fundamental mechanism behind battery charging (ion mobility), catalysis, and phase transitions.

The Microscopic View (Molecular Dynamics):
At finite temperatures, atoms possess kinetic energy and continuously undergo a random walk, colliding and squeezing past their neighbors.

How do we measure it from VASP?
We calculate the **Mean Square Displacement (MSD)**, which tracks how far the average atom drifts from its starting position over time (t).

What is Diffusion? From Macroscopic to Microscopic

The Einstein Relation (3D)

$$D = \frac{1}{6} \lim_{t\to\infty} \frac{d}{dt} \langle |\vec{r}_i(t) - \vec{r}_i(0)|^2 \rangle$$

- The term $\langle |\vec{r}_i(t) - \vec{r}_i(0)|^2 \rangle$ is the MSD.
- In a solid, atoms vibrate in place (MSD is a flat, constant line).
- In a liquid/gas, atoms drift continuously (MSD grows *linearly* with time, and the slope gives us D).

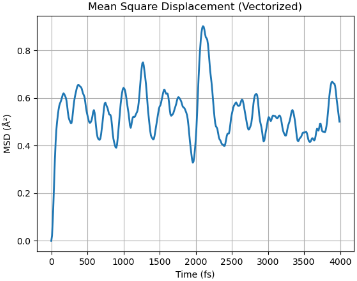
Hands-on: Write Your Own Diffusion Script with AI

Extracting the MSD from a trajectory requires unwrapping the periodic boundary conditions. Instead of writing the math from scratch, we will use an AI assistant to write the Python script for us.

Prompt your AI (ChatGPT/Gemini/Claude)

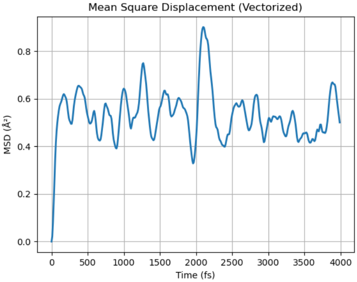
"Write a Python script using the ase library to calculate the Mean Square Displacement (MSD) from a VASP trajectory. Requirements: 1. Read the `XDATCAR` file. 2. Calculate the average MSD for all atoms at each time step. Ensure you unwrap the coordinates to account for periodic boundary conditions. 3. Plot the MSD vs. time (assume a 2 fs time step) using `matplotlib`. 4. Provide clean, well-commented code using only ase, `numpy`, and `matplotlib`."

Diffusion of Li



by running `python diffusion.py` you should get this plot? Is it ok?

Diffusion of Li



by running `python diffusion.py` you should get this plot? Is it ok?

No. Inspect the trajectory, Li didn't melt just oscillates around.

Why? It should melt at 450K. The culprit is a too small supercell; the movement of each atom are too coordinated.

Machine learning force-fields: MLFFs

- We need a larger cell, however, even for a $2 \times 2 \times 2$ we struggled.
- One option is to develop a classical force-field (potential); no electrons.
- VASP can do it *on-the-fly*.
- This option is the safest, as the MLFF gets unreliable, VASP switch back to DFT.

The Danger Zone: When NOT to use MLFFs

Machine Learning Force Fields (MLFFs) are interpolators. They predict energies and forces based on pattern recognition.

1. **The Extrapolation Trap (Unseen Phase Space)**
   - ML models cannot extrapolate. If you train a model on liquid water at 300 K, it will likely fail catastrophically if you use it to simulate boiling at 400 K or ice at 100 K.
   - *Rule of thumb:* Never run an MLFF production run at a temperature or pressure higher than your training data.
2. **Electronic and Magnetic "Blindness"**
   - MLFFs bypass the Schrödinger equation. They throw away the wavefunction to give you raw atomic forces.
   - You cannot extract electronic bandgaps, topological states, or magnetic moments (e.g., spin dynamics, PT symmetry breaking) directly from an ML trajectory.

The Danger Zone: When NOT to use MLFFs

3. **Long-Range Interactions & Charge Transfer**
   - VASP's MLFF relies on *local* descriptors (usually a 5–8 Å cutoff sphere around each atom).
   - They struggle with long-range macroscopic electrostatics (e.g., highly charged ionic liquids) and dramatic redox charge-transfer events where the electronic environment changes abruptly.

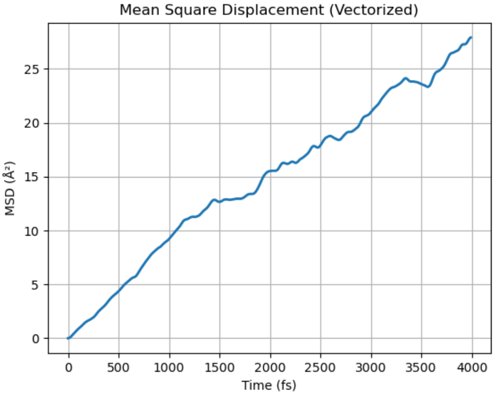
Hands-on 2: MLFF equilibration

- Let's train a MLFF from scratch, this time with a $4 \times 4 \times 4$ cell, 128 atoms.
- The `train` automatically switches between a classical MLFF and DFT, as needed.
- Go to `4_mlff_equilibration` in leftraru, inspect the files and run the calculation.
- use `tail -n run.sh-11075527.out` (use your actual job_id)
- use the **plot.sh**, what is going on? Comment on the sudden jumps.
- Feel free to plot the diffusion, it will be instructive. What is going on?

Hands-on 2: MLFF production

- Let's run a production NVT run using MLFF (training mode)
- go to `5_mlff_production`
- copy your last `CONTCAR` as a `POSCAR`
- copy your last `ML_ABN` to `ML_AB`, so you are using the information of the previous run to build a MLFF.
- inspect the output files and plot the diffusion.

Diffusion of Li at 600K



With the help of MLFF we got a physically sound diffusion, did we?

Diffusion of Li at 600K

With the help of MLFF we got a physically sound diffusion, did we?

- Yes and no. Before 1 ps, yes. After that, the trajectory looks weird.
- Let's look at the `OSZICAR` file. The MLFFs helped to speed-up the calculation during the first 1 ps. After that the MLFFs took over the MD.
- Visualize the trajectory, you will see a tendency for clustering (vaccum regions) inside the crystal
- In general when MLFFs fail, they do it catastrophically.
- **Never blindly trust ML**
- If you remove the memory-saving tags, the diffusion will improve. Anyways, be careful.

Take home message

- Always split your MD into equilibration and production.
- Check that every stage is physically sound.

Do you need MLFFs?

- Really? Not for this example
- Set a comparison against regular AIMD.
- You can control some methodological shortcomings (e.g. cell size), use them to evaluate the MLFF results.